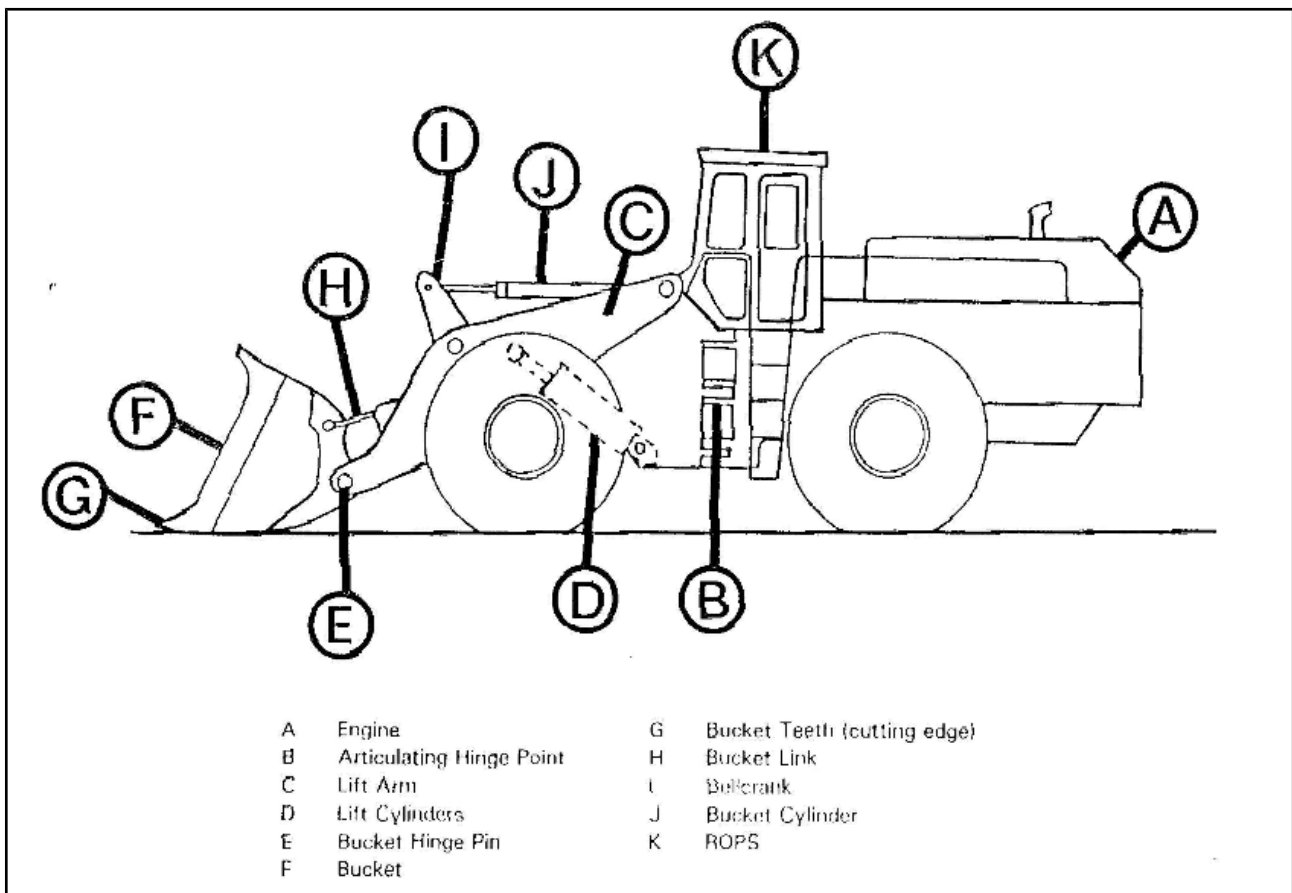


FRONT END LOADERS

Front end loader is a wheel crawler mounted tractor with front mounted bucket. It is utilised in excavating loading and transporting material. Because of its versatility the front end loader is finding a wide variety of mining applications. Front end loader type equipment started as attachments for crawler mounted dozers. At present articulated wheel mounted loaders have also been developed ranging in a capacity of 5 to 24 cubic metres. The design of these units have been defined and is now relatively standardized in its basic configuration. For mining applications the trend has been towards wheel mounted units in sizes more than 5 cubic metre. The larger units for the most part are relatively new to production applications in surface mining. They are however finding used as primary production machines excavating in medium tough conditions and loading up to 170 ton trucks.

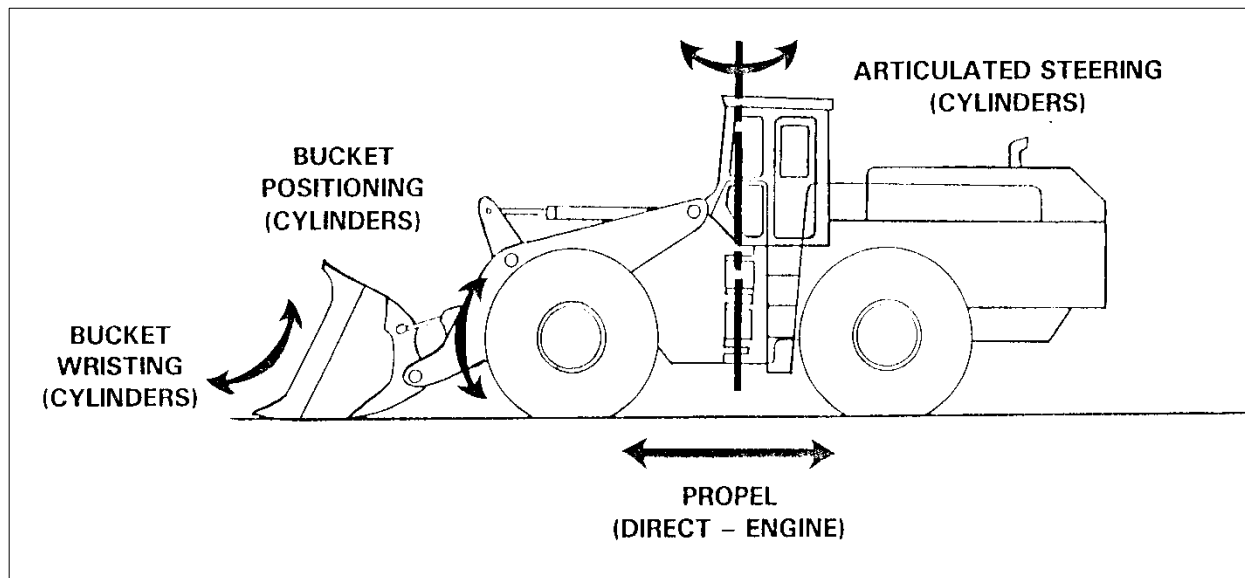


Powered Functions

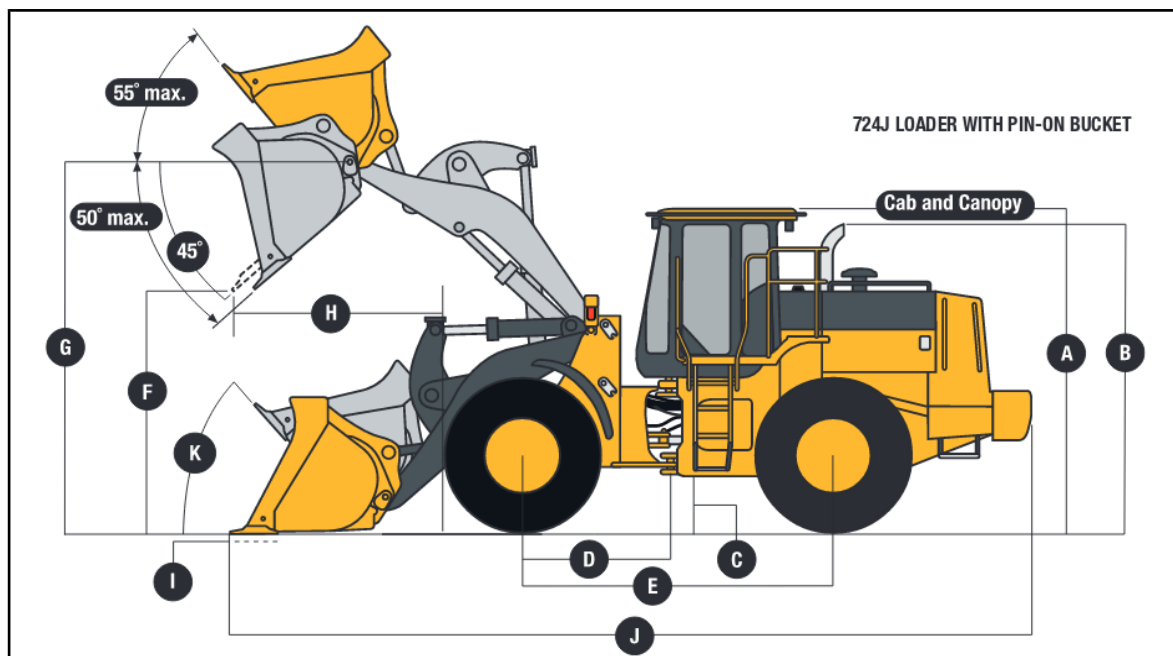
the powered functions of a front end loader are :

- ✓ 4 wheel drive through direct converter or electric wheel motor
- ✓ articulated steering through hydraulic cylinders
- ✓ bucket positioning (raise or lower) through hydraulic cylinders
- ✓ bucket wristing through hydraulic cylinders

the powered functions are explained with the help of following diagram



Representative dimension of a typical Payloader of 3.6 m³ bucket capacity is given below:



A:Height to Top of Cab and Canopy.	11 ft. 2 in. (3412 mm)
B:Height to Top of Exhaust	11 ft. 0 in. (3362 mm)
C:Ground Clearance	17.8 in. (453 mm)
D:Length from Centerline to Front Axle	63 in. (1600 mm)
E:Wheelbase	126 in. (3200 mm)
F:Dump Height , 45 degree dump.	113.5 in (2883 mm)
G:Height to Hinge Pin, Fully Raised.	13 ft. 7 in. (4136 mm)
H:Dump Reach	45.4 in (1152 mm)
I:Maximum Digging Depth.	2.8 in. (71 mm)
J:Overall Length.	26 ft. 11 in. (8195 mm)
K:Maximum Rollback at Ground Level	40 degrees

Applicability

front end loaders can dig/excavate, transport over short distance (generally limited to an economic distance of 50 m) and load by itself. The loading can be into trucks/dumpers, hoppers or belt loaders. Due to these capabilities front end loaders find application both as primary equipment (high capacity ones) and as utility equipment.

Mining applications :

- ◆ dig and load overburden faces where the face height is upto 6 m.
- ◆ Load coal/ore at face
- ◆ dig and load soft material such as fire clay
- ◆ load ore from ore pockets left by big loading machines (scramming)
- ◆ load coal left behind by surface miners in windrow cutting or by side discharging
- ◆ load and/or transport of top soil
- ◆ load from surge piles
- ◆ cleanup tool for shovels and draglines
- ◆ maintain ore/coal stock piles
- ◆ maintain spoil piles
- ◆ land clearing and preparation of location
- ◆ construction and maintenance of access roads and haul roads
- ◆ boost trucks from pit
- ◆ tow disabled vehicles
- ◆ relocate culvert pipes
- ◆ relocate pumps
- ◆ move electric cable skids

Cycle time and Productivity

The cycle time and hourly productivity of front end loader are calculated as given below.

Cycle time = digging time + transport time + dumping time + return time

Digging time depends on the available crowding force of the machine and the resistance offered by the material. Transport and return time depends on the transport distance and the respective load and empty travel speed.

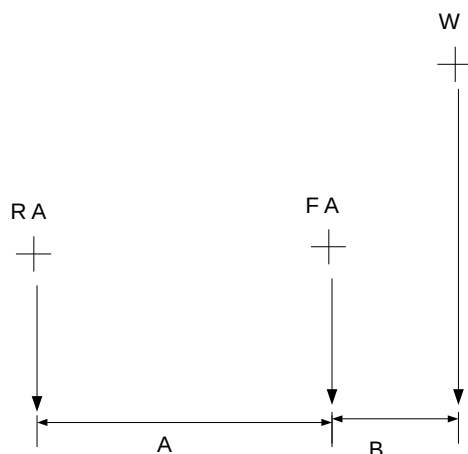
Hourly productivity = Bucket capacity x fill factor x swell factor x bank density x (60/Cycle time in min.) x efficiency

Concept of tipping load

Minimum tipping load is the measure of stability of front end loaders

at minimum tipping load

$$A \times RA = B \times W$$



where,

- RA = machine weight acting along the rear axle
- W = weight of bucket and load
- A = distance between two axles of the machine
- B = the distance between the front axle of the machine and the centre line of the bucket in extended condition

Generally the minimum tipping load is about 70-80% of the machine weight depending on its configuration.

Maximum operating load < 50% of minimum tipping load

the concept of minimum tipping load and maximum operating load is used to find out the suitable bucket size of a particular front end loader for a given job condition.